

Escuela Técnica Superior de
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Analysis Report Do4

Group C1.068 | Diseño y Pruebas II | 05/18/2025

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Repositorio de Github: <https://github.com/DP2-C1-068/Acme-ANS-Do4>

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Resumen ejecutivo

Este documento presenta un informe de análisis centrado en la evaluación y corrección de determinados requisitos del proyecto que requerían aclaración o modificación. El objetivo principal del análisis es asegurar que dichos requisitos se alineen con el comportamiento previsto del sistema, mejorando así la calidad y viabilidad del producto software. Cada registro de análisis incluye una copia literal del requisito original, un análisis detallado de los problemas detectados, las decisiones adoptadas para resolverlos y, cuando corresponde, un enlace a la validación realizada por el docente (foro de la asignatura).

Introducción

El presente informe de análisis documenta el trabajo realizado para aclarar, matizar o corregir ciertos requisitos del proyecto que, durante las primeras fases del diseño del sistema, se identificaron como ambiguos, contradictorios o potencialmente problemáticos para su implementación.

Los registros incluidos recogen, para cada caso, una copia literal del requisito original, un análisis detallado de las dificultades que plantea, las decisiones adoptadas para su resolución, y un enlace a la validación proporcionada por el docente. Este proceso se ha llevado a cabo tanto por correo electrónico con el profesorado como por el foro, mas en este documento solo trataremos los del foro.

Cabe destacar que no se han comentado todos los requisitos del proyecto, sino únicamente aquellos que requerían un análisis adicional, de acuerdo con las directrices establecidas. La metodología seguida ha priorizado la claridad en la argumentación y la justificación de cada decisión tomada.

La estructura de este documento es la siguiente: tras la portada y la tabla de revisiones, se incluye el resumen ejecutivo y esta introducción. A continuación, se presentan los registros de análisis, cada uno de los cuales incluye el requisito original, la pregunta formulada con sus posibles alternativas y sus respectivos pros y contras, la respuesta proporcionada por el profesor y las conclusiones extraídas. Finalmente, se expone una conclusión con las consideraciones finales y una valoración general sobre el estado de los requisitos desde la perspectiva del análisis realizado.

Contenido

Registro de análisis 1

Requisitos involucrados:

[D02-S05-03]

The **technicians** care of aircraft maintenance by conducting regular inspections, performing repairs, and carrying out other maintenance tasks. The system must store the following data about them: a **license number** (unique, pattern " $^{\wedge}[A-Z]\{2-3\}\backslash d\{6\}\$"$), a **phone number** (pattern " $^{\wedge}\backslash +?\backslash d\{6,15\}\$"$), their **specialisation** (up to 50 characters), whether they have passed their **annual health test** or not, and their **years of experience**. Optionally, the system may store their **certifications** (up to 255 characters).

[D02-S05-04]

Maintenance records are comprehensive records of activities performed on a given **aircraft** throughout its operational life. The system must store the following data about them: the **moment** when a maintenance takes place, its **status** ("PENDING", "IN PROGRESS", "COMPLETED"), the next **inspection due date**, an **estimated cost**, and some optional **notes** (up to 255 characters).

[D02-S05-05]

Maintenance records rely on **tasks**. A task is a specific predefined operational duty carried out by a **technician** on **aircrafts**. The system must store the following data about tasks: their **type** ("MAINTENANCE", "INSPECTION", "REPAIR", "SYSTEM CHECK"), a **description** (limited to 255 characters), a **priority** (ranging from 0 to 10), and an **estimated duration** (in hours).

[D02-G-06]

An **aircraft** is a vehicle designed for air travel that belongs to an airline and is used to transport passengers between cities or countries. The system must store the following data about them: a **model** (up to 50 characters), a **registration number** (unique, up to 50 characters), its **capacity** as a number of passengers, its **cargo weight** (between 2K and 50K kgs), a **status**, which reports on whether the aircraft is in active service or under maintenance, and optional **details** (up to 255 characters).

[D02-G-07]

A **service** is an offering provided by airports to enhance a passenger's experience. The system must store the following data about them: a **name** (up to 50 characters), a link to a **picture** that must be stored somewhere else, and its **average dwell time** (in hours); optionally, it can store a **promotion code** (unique, " $^{\wedge}[A-Z]\{4\}-[0-9]\{2\}\$"$, the last two digits correspond to the current year), and the **money** that should be discounted when the promotion code is applied.

Estimado profesor Rafael,

Mi nombre es Gabriel Vacaro, y formo parte del grupo C1-068. me pongo en contacto con usted para solicitar feedback del cliente con respecto a los límites de validación no definidos explícitamente en algunos requisitos, algo muy usual en esta industria.

Contexto:

He realizado una recopilación de los atributos que no están completamente definidos en las **tareas** individuales del Student#5 y los requisitos grupales que causaban confusión, los requisitos con atributos implicados son los siguientes:

D02-S05-03

The **technicians** care of aircraft maintenance by conducting regular inspections, performing repairs, and carrying out other maintenance tasks. The system must store the following data about them: a **license number** (unique, pattern " $^[A-Z]{2-3}\d{6}$$ "), a **phone number** (pattern " $^\d{6,15}$$ "), their **specialisation** (up to 50 characters), whether they have passed their **annual health test** or not, and their **years of experience**. Optionally, the system may store their **certifications** (up to 255 characters).

En este caso, el único atributo que no está bien definido es *years of experience*, si bien el profesor Rafael ha aclarado que debe ser un entero en el siguiente post del foro sobre un requisito similar al descrito pero sobre *Airline managers*:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussion_board&conf_id=426211_1&forum_id=253522_1&message_id=459526_1

y cito textualmente:

Years of experience. Popone Ud. el uso de un número double entre 0.00 y 75.00. El cliente nos indica que tan sólo le interesa saber el número de años en conjunto; no ve diferencia entre 5.1 y 5.9; este valor tan sólo le da una idea aproximada de la cantidad de experiencia acumulada que tiene una persona. Por lo tanto se puede entender como un entero.

No me queda claro que límite superior estima el cliente como adecuado, además en este caso son técnicos, no *mánagers* de aerolíneas, por tanto este límite puede ser diferente incluso estos dos roles.

D02-S05-04

Maintenance records are comprehensive records of activities performed on a given **aircraft** throughout its operational life. The system must store the following data about them: the **moment** when a maintenance takes place, its **status** ("PENDING", "IN PROGRESS", "COMPLETED"), the next **inspection due date**, an **estimated cost**, and some optional **notes** (up to 255 characters).

En este requisito, si bien gracias al hilo del foro comentado anteriormente se han aclarado los atributos de ámbito temporal, el parámetro *estimated cost*, (el cual debe ser de tipo Money, debido a que es una cantidad de dinero) tiene un valor límite máximo muy dependiente del tipo de tareas que se lleven a cabo en el mantenimiento y del avión que se repare, por tanto necesito feedback del cliente para fijar un límite que se ajuste a sus necesidades.

D02-S05-05

Maintenance records rely on **tasks**. A task is a specific predefined operational duty carried out by a **technician** on **aircrafts**. The system must store the following data about tasks: their **type** ("MAINTENANCE", "INSPECTION", "REPAIR", "SYSTEM CHECK"), a **description** (limited to 255 characters), a **priority** (ranging from 0 to 10), and an **estimated duration** (in hours).

En este requisito, hay dos atributos implicados, por una parte, *estimated duration*, de nuevo al igual que en los anteriores, si bien debe ser un entero debido a que es una estimación imprecisa, necesito aceptación del cliente para establecer un límite superior conveniente.

Por otra parte, *priority*, que si bien quedan definidos los límites superior e inferior del rango, no lo hace la precisión de la puntuación. ¿Debe ser un entero o un real? ¿en caso de ser real, necesitará una cifra decimal o dos?

D02-G-06

An **aircraft** is a vehicle designed for air travel that belongs to an airline and is used to transport passengers between cities or countries. The system must store the following data about them: a **model** (up to 50 characters), a **registration number** (unique, up to 50 characters), its **capacity** as a number of passengers, its **cargo weight** (between 2K and 50K kgs), a **status**, which reports on whether the aircraft is in active service or under maintenance, and optional **details** (up to 255 characters).

En este requisito, hay un atributo implicado: *capacity*, claramente la capacidad de una aeronave variará en gran medida, mas no cuento con la información de que tipo de aviones necesitará albergar el sistema, y por tanto me veo en la obligación de preguntar al cliente por este límite superior.

D02-G-07

A **service** is an offering provided by airports to enhance a passenger's experience. The system must store the following data about them: a **name** (up to 50 characters), a link to a **picture** that must be stored somewhere else, and its **average dwell time** (in hours); optionally, it can store a **promotion code** (unique, "[^][A-Z]{4}-[0-9]{2}\$", the last two digits correspond to the current year), and the **money** that should be discounted when the promotion code is applied.

El único atributo implicado en este requisito es *discountedMoney*, no quedándome claro cuál debería ser el límite inferior y superior. Si bien en el hilo mencionado se comenta que un vuelo puede valer entre 0.00 XXX y 1,000,000.00 XXX, siendo XXX la divisa, no sabría que porcentaje podría ser un descuento razonable a ojos del cliente.

Alternativas propuestas:

1. Elegir de forma informada límites razonables para cada atributo de cada requisito:

- **Ventajas:** No se observan ventajas.
- **Desventajas:** Seguramente haya discrepancias entre lo que pide el cliente y lo prefijado sin consultarlo. Nadie sabe mejor que el cliente que es lo que necesita su negocio, y por tanto no es buena idea fijar límites de forma arbitraria sin consultárselo.

2. Consular con el cliente:

- **Ventajas:** Recibimiento de feedback directo, no damos por hecho lo que quiere el cliente, esperamos que nos lo diga y así nos aseguramos no equivocarnos.
- **Desventajas:** no se observan desventajas.

3. Contactar con el cliente con una serie de valores propuestos y preguntarle si le parecen correctos:

- **Ventajas:** Recibimiento de feedback directo, no damos por hecho lo que quiere el cliente, esperamos que nos lo diga y así nos aseguramos no equivocarnos. Además facilitamos la selección de valores al cliente y damos una buena imagen de preparación y trabajo.
- **Desventajas:** No se observan desventajas.

Solución propuesta:

Hemos optado por la **tercera alternativa**, los valores que le proponemos al cliente como límites son los siguientes:

Years of experience (technician): Entero entre 0 y 50. (es un trabajo físico con gran desgaste)

estimated cost (maintenance record): Money entre 0.00 XXX y 999,999.00 XXX, siendo XXX la divisa. (Los aviones de gran tamaño pueden llegar a requerir de piezas o reparaciones muy caras)

estimated duration (task): Entero, de 0 a 100 (existen tareas como cambio de motores que, en grandes aeronaves requieren de gran cantidad de tiempo)

priority (task): Entero, de 0 a 10. (estimo que la precisión de la importancia no necesita decimales).

capacity (aircraft): Entero, de 1 a 250 (estimación realizada según algunas búsquedas en modelos que soportan 50k kgs de capacidad de almacenaje máximo, que la máxima capacidad del atributo **cargo weight**)

discountedMoney (Service): Money, de 0.00 XXX a 1,000,000.00 XXX (desde el vuelo más barato al más caro).

Espero me comente que le parecen estos valroes límite y nos aconseje cuáles son los que Ud. ve más correctos.

Gracias por su atención.

Saludos, GV

Estimado Gabriel María:

Tenemos varios mensajes sobre el mismo tema. Veo que se centra Ud. en los requisitos del estudiante S05. En los mensajes [1, 2, 3, 4] podrá encontrar Ud. una argumentación en relación al porqué no es una buena idea mandar varios mensajes sobre el mismo tema: por una parte tenemos la percepción del cliente y por otra la dificultad de gestionar varias respuestas a un mismo tema.

En el mensaje [4], le propuse al estudiante que lo envió que liderara el proceso para recopilar una lista exhaustiva de entidades/atributos que necesitan aclarar su dominio. Creo que en este mensaje encontrará una buena lista en relación al estudiante S05 y en la respuesta a [1] encontrará una lista en relación al estudiante S01.

Como hemos comentado en varias ocasiones, aclarar los rangos de los tipos de datos es algo que se olvida habitualmente en los documentos de requisitos porque para el cliente esto no es lo importante de su negocio y para el ingeniero de requisitos son muchos; así que es normal que las dudas surjan cuando se va a implementar el modelo de datos o a preparar datos de pruebas (ahí está bien porque estamos al comienzo del proyecto y aún podemos hacer cambios fácilmente) o cuando se preparan las pruebas (ahí no está tan bien porque ya estamos avanzados en el proyecto y hacer cambios no es tan sencillo, aparte de la mala sensación que puede provocar en el cliente que cuando estamos llegando al final le preguntemos sobre algo que incumbe a los datos propiamente dichos).

Lo que ocurre es que tenemos que buscar una forma más efectiva de resolver este problema que va a ser tan común. Escribir párrafos individuales por cada entidad/atributo puede resultar bastante farrogoso, no sólo de escribir sino después de entender cuando tengamos que usar esta información. En estos casos, una tabla con las siguientes columnas es mucho más efectiva: entidad/atributo, rango-propuesto, comentario-equipo-desarrollo, validación-cliente, comentario-cliente. Con estas columnas podemos fácilmente montar una tabla en la que va a resultar muy sencillo identificar qué falta, hacer propuestas y que el cliente las valide o las cambie.

Saludos. RC

[1] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion board&conf id= 426211_1&forum id= 253522_1&message id= 459526_1

[2] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion board&conf id= 426211_1&forum id= 253522_1&message id= 460252_1

[3] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 460339_1

[4] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&conf_id= 426211_1&forum_id= 253522_1&message_id= 460574_1&thread_id=460574&nav=discussion_board&course_id= 89154_1

Lecciones aprendidas del primer análisis:

- Se ha aprendido la importancia de aunar las incompletitudes y dudas sobre los valores límite de cada entidad para no sobrecargar al cliente y dar una mala impresión. En este caso en concreto, un grupo sobresalió sobre el resto proporcionando un libro de Excel con todos los atributos de todas las entidades del proyecto.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 460729_1

Registro de análisis 2

Requisitos involucrados:

[D02-S05-4]

Maintenance records are comprehensive records of activities performed on a given **aircraft** throughout its operational life. The system must store the following data about them: the **moment** when a maintenance takes place, its **status** ("PENDING", "IN PROGRESS", "COMPLETED"), the next **inspection due date**, an **estimated cost**, and some optional **notes** (up to 255 characters).

Estimado profesor Rafael,

Mi nombre es Gabriel Vacaro, y formo parte del grupo C1-068. me pongo en contacto con usted para solicitar feedback del cliente con respecto a la validacion custom del requisito [D02-S05-4].

Contexto:

El requisito dicta de la siguiente forma:

Maintenance records are comprehensive records of activities performed on a given **aircraft** throughout its operational life. The system must store the following data about them: the **moment** when a maintenance takes place, its **status** ("PENDING", "IN PROGRESS", "COMPLETED"), the next **inspection due date**, an **estimated cost**, and some optional **notes** (up to 255 characters).

El problema surge cuando al interpretar el requisito, podemos deducir que el momento en el que empieza el mantenimiento sucederá en una fecha anterior a la fecha de la próxima inspección, pues por definición, hasta que no acabe la inspección que se esta realizando, no podrá empezar la siguiente.

Alternativas propuestas:

- A1: No crear validación custom y permitir que haya "sigueintes inspecciones" con fecha anterior a la fecha de inicio de mantenimientos actuales:
 - ventajas: No se observan ventajas.

- desventajas: se permite la inclusión de datos inconsistentes (futuras inspecciones en fechas anteriores al comienzo del mantenimiento actual).
- A2: Crear validación custom y obligar a que la siguiente fecha de inspección sea al menos 1 minuto posterior a la fecha de comienzo del mantenimiento actual.
 - ventajas: se elimina esta inconsistencia en los datos.
 - desventajas: No se observan desventajas.

Solución electa:

Elige la A2, debido a que no encuentro desventajas relativas a implementar esta restricción custom, pero si encuentro situaciones en las que implementarla solventaría posibles problemas de consistencia en los ficheros de población.

Gracias de antemano.

Saludos, GV.

Estimado Gabriel María:

Gracias por seguir los consejos que les proporcionamos, incluido el de identificar en el título el requisito al que hacen referencia, dado que esto facilita localizar los mensajes referidos al mismo.

Su duda es sobre si el momento (moment) en que un mantenimiento tiene lugar y la fecha de próxima inspección (inspection due date) tienen alguna restricción que satisfacer. Por favor, recuerde que los requisitos se enuncian linealmente, pero no se pueden interpretar más que cuando se leen todos y se entienden sus relaciones. En especial, es muy recomendable estudiar siempre de forma conjunta los requisitos de información y los funcionales, al menos. En este caso es importante tener en cuenta el requisito D03-S05-08, que indica lo siguiente:

[D03-S05-08] Operations by technicians on maintenance records: - List their maintenance records. - Show the details of the maintenance records, including the tasks involved. - Create, update, and publish maintenance records. Please, note that to publish a maintenance record, it cannot have any unpublished tasks and should have at least one published task.

Le aclaro que el momento en que un “mantenimiento tiene lugar” (maintenance takes place) es el momento en que el técnico correspondiente registra que debe realizar ese mantenimiento (status = PENDING), que ya ha empezado a trabajar en el mismo (status = IN PROGRESS), o que ya lo ha terminado (status = COMPLETED). Dicho en otras palabras: ese atributo “moment” registra el momento de creación, de actualización o de publicación de un registro de mantenimiento; es decir, se actualiza cada vez que un técnico edita un registro de mantenimiento.

Ahora, la cuestión es qué significa exactamente “próxima inspección”, dado que es una expresión nueva a la que tan sólo se hace referencia en este requisito. Se consulta con el cliente y nos aclara que tanto “maintenance” como “inspection” hacen referencia a lo mismo: operaciones que los técnicos realizan sobre los aviones para garantizar su correcto funcionamiento; en unas ocasiones sólo llevan consigo realizar comprobaciones para certificar que todo está bien y en ese caso se suele hablar de “inspections”; pero en otras ocasiones es necesario hacer reparaciones y en ese caso se suele hablar de “maintenance”. Por lo tanto se trata de sinónimos que usa nuestro cliente y debemos adaptarnos a su vocabulario y preservarlo en nuestro proyecto tan bien como sea posible. El cliente también aclara que al hablar de “próxima fecha de inspección” (next inspection due date), se refiere a una fecha que el propio técnico registra como apropiada para realizar el siguiente mantenimiento, aunque esto se toma simplemente como una sugerencia por parte del técnico, no como una obligación.

Por lo tanto, parece que tiene sentido que la próxima fecha de inspección sea posterior al atributo “moment” que indica el último momento de actualización. En principio, la fecha se puede fijar en cualquier momento, pero,

como indica el requisito funcional asociado, siempre y cuando un registro de mantenimiento no esté ya publicado.

¿Podríamos adelantar algún análisis con relación al requisito funcional D03-S05-08 teniendo en cuenta lo que acabamos de comentar? Si podamos ir adelantando ese análisis, perfecto, porque es justamente ahora cuando más fresco tenemos los requisitos de información relacionados con el mismo.

En relación con sus alternativas:

- A1: no parece la adecuada dado que el propio cliente no ha aclarado que la próxima fecha de inspección es una sugerencia de los técnicos y debe fijarse después del atributo "moment".
- A2: teniendo en cuenta todo lo comentado, parece la más lógica.

Por lo tanto, la solución que propone (A2) parece la más adecuada en el contexto de Acme ANS y los requisitos del estudiante S05. Tan sólo una puntualización: por favor, recuerde que para poder hacer pruebas repetibles necesitamos trabajar con un reloj simulado en las plataformas de desarrollo y testing, pero con un reloj real en la de explotación. Recuerde que el framework proporciona la clase de utilidad "MomentHelper", en la que puede encontrar métodos que le permitirán realizar comparaciones temporales teniendo en cuenta el detalle anterior. Podrá encontrar muchos ejemplos de uso en el proyecto Acme Jobs.

Saludos. RC

Estimado profesor Rafael:

Le respondo de nuevo en este hilo debido a que, tras realizar la custom validation mencionada en los mensajes anteriores, me surge una nueva duda con respecto a la población de la entidad MaintenanceRecord del requisito **[D02-S05-6]:**

Produce assorted sample data to test your application informally. The data must include two **technician** accounts with credentials "**technician1/ technician1**" and "**technician2/ technician2**". Create an additional technician account with credentials "**technician3/ technician3**" that represents a technician with no data, but his or her profile.

Contexto

Tras la realización de la validacion anteriormente comentada (que el atributo "inspectionDueDate" sea al menos un minuto posterior al atributo "moment") me encuentro con varias dudas al respecto de los valores límite de estos atributos. Como se ha comentado ya en varios hilos, los límites de estos atributos deben ser los establecidos por defecto en el sistema; en este caso, el momento mínimo es: 2000/01/01 00:00, y el máximo: 2201/01/01 00:00.

El problema nace cuando, según hemos comentado, el atributo "inspectionDueDate" debe ser al menos un minuto posterior al atributo "moment", lo que genera ciertas restricciones a la hora de generar datos de prueba, como podemos observar en la siguiente tabla:

MaintenanceRecord:Moment	MaintenanceRecord:InspectionDueDate	Notes
?	2000/01/01 00:00	Inconsistencia, preceda a 2000 antelación

2201/01/01 00:00	?	Inconsistencia, 2201/01/01 00:00
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Donde "?" significa que cualquier valor es incorrecto.

Cuestion

¿Cómo probar los límites de estos atributos?

Alternativas propuestas:

- **A1:** Probar los valores límite que no generen incongruencias (todos menos los dos descritos en la tabla)
 - ventajas: Respeta la decisión de nuestro cliente de dejar los límites en default.
 - desventajas: se quedan sin probar dos valores límite que si estan especificados como válidos en los límites pero son inalcanzables.
- **A2:** Utilizar los parámetros "min" y "max" de la anotacion "@ValidMoment" en los atributos "moment" e "inspectionDueDate" para limitar el valor de "moment" (del 2000/01/01 00:00 al 2200/12/31 23:59) y el valor de "inspectionDueDate"(del 2000/01/01 00:01 al 2201/01/01 00:00). Generando así nuevos valores límite que sí pueden ser probados.
 - **ventajas:** Es una implementación que no deja lugar a incongruencias.
 - **desventajas:** No respeta estrictamente la indicación del cliente.

Alternativas electa:

Elijo la alternativa **A2**, dado que de esta forma somos capaces de probar todos los valores límite y el sistema queda exento de posibles malentendidos en el futuro (véase por ejemplo el caso de creación de un maintenanceRecord, donde el usuario fija el atributo "moment" a 2201/01/01 00:00, lo cual sería válido, pero cualquier valor introducido en "inspectionDueDate" resultaría en un error de validación).

Gracias de antemano.

Saludos, GV.

Estimado Gabriel María:

La conclusión a la que ha llegado Ud. es que hay datos de prueba correctos (se les suele llamar positivos) e incorrectos (se les suele llamar negativos).

Ahora mismo estamos en la L02 y aquí nuestra tarea es preparar un conjunto de datos de ejemplo (sample data) que nos permitan cargar en la BD información que posteriormente no será de utilidad para realizar pruebas; evidentemente, en este momento sólo podemos generar y grabar datos correctos (positivos).

Los datos incorrectos (negativos) los usaremos en la lección L03 para realizar pruebas informales y en la L04 para realizar pruebas formales. Tendremos que comprobar que nuestro sistema acepta todos los datos positivos y rechaza todos los negativos.

Por lo tanto, ahora mismo no podemos darle ninguna utilidad a los datos negativos.

Fíjese en que el fichero helper “sample-data.xlsx” incluye unas celdas en color naranja bajo el epígrafe “CLAZZ” cuyos valores son “+” o “-”. Intuitivamente, el “+” le indica que el dato generado *podría* ser positivo (correcto, lo podemos usar en L02) y el “-” que *podría* ser incorrecto (negativo, para la L03 o L04). Fíjese en el énfasis en “podría” dado que el helper no es más que una hoja de cálculo que nos facilita un poco el trabajo para generar datos de prueba, pero no tiene información alguna sobre los requisitos; necesita usar su NI para entender los requisitos y finalmente dilucidar si un dato se puede considerar positivo o negativo.

Fíjese también en que este helper tan sólo le da una ayuda con los atributos de forma individual. ¿No sería una buena idea emplear algo de tiempo en extenderla de forma que pueda ayudarle aún más en el caso de atributos interrelacionados? Por ejemplo, ese momento de actualización y ese momento de próxima inspección. Le dejo a su criterio que valore el beneficio/coste de realizar esa mejora.

Con respecto a sus alternativas, le comento:

- A1: lo que llama “incongruencias” es lo que habitualmente se denomina “positivo” o “negativo”. La desventaja que Ud. menciona no es tal; se realizarán pruebas completas con datos negativos, pero para ello tenemos que esperar a la L03 y L04. En la L02 sólo podemos trabajar con datos que podamos meter en la BD y esos datos deben ser positivos.
- A2: no se lo recomiendo. Lo normal en el caso de los momentos y las cantidades de dinero es que se puedan definir límites globales a nivel de aplicación. Por favor, tenga en cuenta que el hecho de que exista un límite no significa que los valores dentro de esos límites tengan sentido. Por ejemplo, es posible que más de un estudiante piense que nadie puede tener 120 años de experiencia profesional. Seguramente no, pero el límite no significa eso. El límite explícito que debemos establecer debe garantizar que se pueden almacenar todos los datos con sentido y que podemos hacer pruebas, no que un valor sea posible o no. En un contexto profesional avanzado se suele distinguir entre el concepto de hard constraint y el de soft constraint; las primeras son restricciones de obligado cumplimiento, mientras que las segundas dan lugar a avisos. Pero en el contexto de la asignatura, no ponemos ninguna restricción soft; recuerde que es un proyecto realista, no un proyecto real y mucho menos avanzado.

Por lo tanto, la alternativa A2 no es adecuada en este caso. Espero que las explicaciones anteriores le hayan aclarado el motivo.

Saludos. RC

Lecciones aprendidas:

- Es importante entender que en fases tempranas del desarrollo solo se pueden usar datos válidos y positivos, dejando los casos negativos para etapas posteriores de prueba.
- No se deben imponer restricciones arbitrarias basadas en suposiciones subjetivas; los límites deben permitir almacenar todos los datos con sentido y facilitar la realización de pruebas.
- En proyectos educativos, conviene centrarse en restricciones técnicas esenciales (hard constraints) y no en validaciones avanzadas (soft constraints), reservadas para contextos profesionales más complejos.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 460922_1

Registro de análisis 3

Requisitos involucrados:

[D03-S05-8]

Operations by **technicians** on **maintenance records**:

- List their maintenance records.
- Show the details of the maintenance records, including the tasks involved.
- Create, update, and publish maintenance records. Please, note that to publish a maintenance record, it cannot have any unpublished tasks and should have at least one published task.

[D03-S05-9]

Operations by **technicians** on **tasks**:

- List their tasks and show their details.
- Create, update, delete, and publish tasks. Note that published tasks cannot be updated or deleted.

Dear Professor Rafael:

I am Gabriel Vacaro, from group C1.068 of the Design & Testing II subject, and since it is strongly recommended, I will try to write my questions in English from now on.

I am contacting you because I have a doubt regarding the implementation of the following requirements (They are very similar to the mandatory D03 requirements of the other students, so I hope this question helps):

[D03-S05-8]

1. Operations by **technicians** on **maintenance records**:

- List their maintenance records.
- Show the details of the maintenance records, including the tasks involved.
- Create, update, and publish maintenance records. Please, note that to publish a maintenance record, it cannot have any unpublished tasks and should have at least one published task.

[D03-S05-9]

2. Operations by **technicians** on **tasks**:

- List their tasks and show their details.
- Create, update, delete, and publish tasks. Note that published tasks cannot be updated or deleted.

Context

The requirement states that technicians must be able to "List their maintenance records" and "List their tasks and show their details". The thing is, in the context of Acme-ANS, the tasks are a catalogue composed of all tasks created by technicians, regardless of the creator. So it would make more sense to be able to list the full catalogue of tasks (ListService) instead of the part of the catalogue that is owned by the principal (List-mineService).

In the case of maintenance records it is clearer to me, because the principal may only want to check his maintenance records in order to work on them, not all technicians' maintenance records. (implement only List-mineService)

Therefore, my questions are:

Questions

When the requirement establishes that technicians must be able to "List their tasks", is it referring to the full catalogue of tasks, the ones that the principal has created or both in different services (List-mineService and ListService/List-allService)?

When the requirement establishes that the technicians must be able to "List their maintenance records", is it referring to all maintenance records, the ones the principal has created or both in different services (List-mineService and ListService/List-allService)?

Here, List-mineService indicates that the technician will only be able to see their creations (regardless of draft mode). And ListService means that the principal will be able to see all **published** tasks/maintenanceRecords.

Alternatives considered

- **Alternative 1:** Implement *List-mineService* for tasks and maintenance records and *ListService* for tasks.
 - **Pros:** The principal may want to see the tasks that they have created in isolation and the full catalogue, but the client may not want the principal to see the full maintenance record catalogue.

- **Cons:** Lacking a functionality that the client may want but may not be in the requirement due to incomplete annotation from the Requirement Engineer.
- **Alternative 2:** Implement *ListService* for tasks and *List-mineService* for maintenance records.
 - **Pros:** I see no pros.
 - **Cons:** Lacking two functionalities that the client may want but may not be in the requirement due to incomplete annotation from the Requirement Engineer.
- **Alternative 3:** Implement *ListService* and *List-mineService* for both tasks and maintenance records.
 - **Pros:** It is the most complete alternative.
 - **Cons:** We may be exceeding the requirement and implementing things that the client did not ask for.

Proposed solution

I chose the third alternative, in which the principal would be able to see all the published tasks in the catalogue (TechnicianTaskListService), his own tasks in isolation (published and drafts) (TechnicianTaskList-mineService), all the published maintenance records (TechnicianMaintenanceRecordListService), and his maintenance records (published and drafts) (TechnicianMaintenanceRecordList-mineService).

I made this decision because the requirements are read linearly but understood globally, so when the client asks for publishing a task or a maintenance record, it may be for the rest to access it, and the best way to access this information is through a listing service.

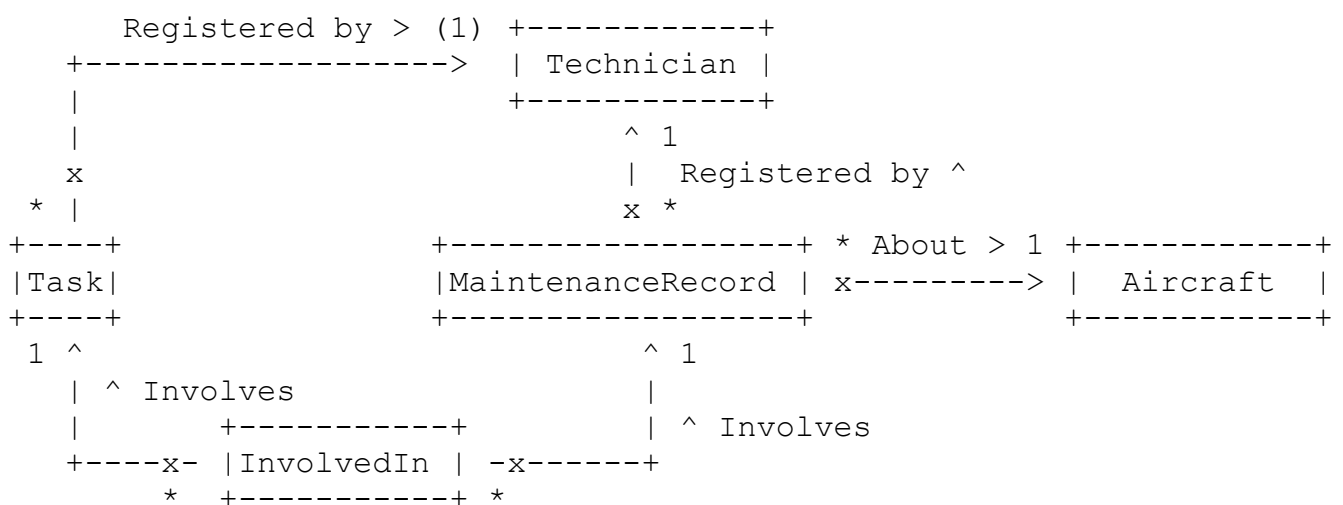
Thanks in advance.

Regards, GV

Dear Gabriel María,

Thanks for taking our pieces of advice into account. Just a quick note; you mention that “[My requirements] are very similar to the mandatory D03 requirements of the other students”. That’s true, but there are no two equal requirements; each student must take the details in his or her requirements into account and the solutions are slightly different.

Your question is about requirements D03-S05-08 and D03-S05-09, which basically report on the operations that technicians have to perform on maintenance records and tasks. The far, the model involved seems to be something like this [1]:



You mention that “the tasks are a catalogue composed of all tasks created by technicians, regardless of the creator.” That is true [3]. Please recall that a “task” is a concept, something like “chacek the battery”; the tasks are somewhat instantiated in maintenance records: when a technician records that he or she’s performed a particular task on a particular aircraft. As you mention, the tasks are organised into sort of a catalogue that is maintained by all technicians. That means that:

- A task that is being edited by a technician (in draft mode) must be kept secret. Only the technician who is editing it can see it, update it, delete it, or publish it. Note that requirement D03-S05-09 explicitly

states that tasks must be published, which means that they are in draft mode as long as they are not published.

- Once a task has been published, it cannot be modified, but it makes sense that all technicians can list it.

Thus, it's a bit odd that Requirement D03-S05-09 indicates that technicians "can list their tasks", which seems to indicate that they cannot list tasks by other technicians. The customer is contacted to clarify this issue and the conclusion is that Requirement D03-S05-09 should be written like this:

[D03-S05-09] Operations by technicians on tasks: - List **the tasks that are published** and show their details. - Create, Update, delete, and publish **their** tasks. Note that published tasks cannot be updated or delete.

It's also a bit off that Requirement D03-S05-08 indicates that technicians "List their maintenance records". It seems that they are sort of information that all technicians must share, since it does not make a lot of sense that a technician checks the battery today and another technician checks it tomorrow. The customer is contacted to clarify this issue and the conclusion is that Requirement D03-S05-08 should have been drafted as follows:

[D03-S05-08] Operations by technicians on maintenance records: - List the **maintenance records that are published** and show their details. - Create, Update, delete, and publish **their** tasks. To publish a maintenance record, all of the tasks involved (at least one) must have been published previously. Note that published maintenance records cannot be updated or deleted.

Regarding your questions:

- Q1: Note the change in the requirement: a technician must be able to list the tasks that are published and show their details, independently from whether he or she created it or not. Obviously, a task in draft mode can only be displayed or edited by the technician who created it. Regarding the name of the feature, it can be "technician/task/list" would work when returning all published tasks. But, you may easily think of an additional feature like "technician/task/list-mine" to retrieve only the tasks that the principal has created.
- Q2: Note the change in the requirement: a technician must be able to list the maintenance records that are published and show their details, independently from whether he or she created it or not. Obviously, a maintenance in draft mode can only be displayed or edited by the technician who created it. Regarding the name of the corresponding feature, something like "technician/maintenance-record/list" would work when listing all published maintenance records, but you can easily implement an additional feature that lists only the maintenance records that correspond to the principal who requests it. Note that implementing something like "technician/maintenance-record/list-mine" would work well and would be the easiest approach (This approach is used in Acme Jobs, for instance).

Note that the idea of implementing the "list-mine" features is a simple and effective approach to the problem. It's conceptually clear and makes a clear separation between both features. That's the preferred approach in Acme Jobs, for instance. But you might easily think of features like "technician/maintenance-record/list?mine=true" or "technician/task/list?mine=true", where the presence of the Boolean parameter "mine" in the URL determines whether the list must retrieve all published entities, or only the entities created by the principal. Please, think of some pros and cons regarding this alternative before making a decision.

Regarding your alternatives:

- A1: note the change in the requirements. The customer has made it explicit which tasks/maintenance records a technician can list and show and which ones he or she can edit.
- A2: same as before.
- A3: that's definitely the right alternative, but, please, read my previous comment: implementing two independent features is OK; no significant problem with that. But what about introducing that "mine" parameter in the URL?

Regards, RC

- [1] https://ev.us.es/webapps/discussionboard/do/message?conf_id=426211_1&forum_id=253522_1&course_id=89154_1&action=list_messages&nav=discussion_board&message_id=461830_1
- [2] https://ev.us.es/webapps/discussionboard/do/message?conf_id=426211_1&forum_id=253522_1&course_id=89154_1&action=list_messages&nav=discussion_board&message_id=461415_1
- [3] https://ev.us.es/webapps/discussionboard/do/message?conf_id=426211_1&forum_id=253522_1&course_id=89154_1&action=list_messages&nav=discussion_board&message_id=461415_1

Dear RC,

After carefully considering and analyzing the pros and cons of the proposed alternatives, I have reached the following conclusion:

Pros and Cons of Each Approach

1. Separate Endpoints (list-mine and list)

Pros:

- Clear separation of functionalities, making it explicit whether a technician is viewing their own records or the entire catalog.
- Simpler to implement and maintain, as each endpoint has a well-defined responsibility.
- Follows a widely used design pattern, improving code readability and maintainability.

Cons:

- Much more redundant, as two endpoints are required for similar functionalities.
- If additional filtering criteria are needed in the future, this approach might require adding more endpoints (not very scalable).
- As engineers, we should aim to reuse and abstract common code blocks. This could be a good opportunity to do so.

2. Single Endpoint with Boolean Parameter (list?mine=true/false)

Pros:

- Reduces endpoint redundancy, making the WIS more concise.
- Provides greater flexibility, as additional parameters could be added easily in the future via URL.
- Allows dynamic filtering without needing multiple separate services and adding lines to the controller.

Cons:

- Slightly more complex to implement, as it requires handling conditional structures (if/switch) in load method of the service.
- Might lead to confusion if not well-documented, as users must understand the meaning of the mine parameter.
- Can introduce additional processing overhead if not properly optimized.

Final Decision

After weighing these factors, I have decided to proceed with **Option 2** (a single endpoint with a boolean parameter). The main reason is its flexibility and scalability, which will allow for easier future expansions while keeping the WIS structure clean.

Thank you for your guidance and insights. I look forward to your feedback.

Best regards, GV

Dear Gabriel María,

Thanks for addressing my questions, analysing pros and cons, and making a decision. Your message is related to implementing two separate endpoints that allow technicians list his or her own tasks ("technician/task/list-mine") or a single endpoint with a parameter that indicates whether the listing must list

his or her own tasks or all published tasks ("technician/task/list?mine=true" or "technician/task/list?mine=false").

Regarding your cons, you comment on "redundancy" but the problem here is not actually "redundancy", but some duplication of code. In general, duplication of code is not a good practice, but it's not such a huge problem in a real-world software project that involves more than a couple of developers. In general, it's a good idea to focus on your customer's requirements first since not fulfilling them is an actual huge problem.

In principle, both alternatives will be accepted in this subject as long as they are correctly implemented. There's only one additional glitch: realise that a technician can list tasks in three different contexts: display his personal catalogue of tasks (which includes both published and unpublished tasks, that is "list?mine=true"), display the full catalogue of tasks (which includes only published tasks since being able to display a task in draft model by another technician might easily be considered a hacking attempt, that is "list?mine=false", and display the listing of tasks that are involved in a particular maintenance record. The last context requires a different parameter, e.g., "list?maintenanceRecordId=999", or a different feature, e.g., "/technician/maintenance-record/list-tasks?id=999".

That said, solutions in which a student implements three endpoints will be accepted (since it enhances clarity and separation of concerns); solutions in which a student implements a single endpoint whose semantics are controlled by means of a parameters will also be accepted (since it requires less code to write independent features and it is not too difficult to implement, but definitely your logic to decide on the buttons in that listing will be a little more complex).

Regards, RC

Lecciones aprendidas:

- Los requisitos deben leerse con atención, pero también interpretarse con una visión global, ya que a veces la intención del cliente no queda perfectamente reflejada en el texto inicial y pueden requerirse aclaraciones posteriores.
- Es fundamental distinguir entre los datos públicos (publicados) y los privados (en borrador): solo el técnico autor puede ver y modificar sus borradores, mientras que los elementos publicados deben estar disponibles para todos los técnicos.
- Existen varias formas válidas de implementar los servicios de listado, ya sea mediante endpoints separados (List y List-mine) o con un único endpoint con parámetros (List-mine=true/false), y la elección debe basarse en la claridad, escalabilidad y mantenimiento del sistema.
- Elegir entre duplicación de endpoints o parametrización implica sopesar claridad frente a reutilización de código, pero lo más importante es cumplir correctamente con los requisitos del cliente, incluso si eso implica un poco más de código duplicado.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf id= 426211_1&forum id= 253522_1&message id= 462839_1

Registro de análisis 4

Requisitos involucrados:

[D03-S05-08]

Operations by **technicians** on **maintenance records**:

- List their maintenance records.
- Show the details of the maintenance records, including the tasks involved.
- Create, update, and publish maintenance records. Please, note that to publish a maintenance record, it cannot have any unpublished tasks and should have at least one published task.

Dear Professor Rafael:

I hope this post finds you well.

My name is Gabriel Vacaro, from group C1.068 of the Design & Testing II subject.

My name is Gabriel Vacaro, and I am a student in group C1.068 for the **Design & Testing II** course. I am reaching out to you because I have a question regarding the implementation of the following requirement:

[D03-S05-8]

1. Operations by **technicians** on **maintenance records**:

- List their maintenance records.
- Show the details of the maintenance records, including the tasks involved.
- Create, update, and publish maintenance records. Please, note that to publish a maintenance record, it cannot have any unpublished tasks and should have at least one published task.

Context

The requirement states that technicians must be able to "list, show, create, update, and publish" maintenance records. However, in the context of **Acme-ANS**, it would make sense to allow technicians to **delete** a maintenance record as long as it has not been published. If a maintenance record is deleted, the tasks associated with it would also be removed from the intermediate table **"Involves"**, but the tasks themselves would remain unchanged.

I have already implemented this service, but I am encountering an issue when deleting a maintenance record under a specific set of conditions. While the service functions correctly under normal circumstances, following these steps leads to an authorization error from show service:

- Log in as **technician1**.
- Navigate to **[List my maintenance records]** (<http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/list?mine=true>).
- Click on any **unpublished** maintenance record (e.g., <http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/show?id=163>).
- Click on the **[Tasks]** button to view the maintenance record's associated tasks. **If you access this view directly via the URL instead** (<http://localhost:8080/Acme-ANS-D03/technician/task/list?maintenanceRecordId=163>), **the bug does not occur**.
- Return to the maintenance record's show view, **but instead of** clicking the "Return" button within the application or navigating directly to the URL (<http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/show?id=163>), use the **browser's back button** (the left arrow in Firefox, located to the left of the URL bar and the refresh button).
- Click **[Delete]**.

At this point, the maintenance record is deleted, but instead of being redirected to:

<http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/list?mine=true>

the system attempts to redirect to:

<http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/show?id=163>

Since the maintenance record no longer exists, the authorization check fails, leading to an error.

Questions

- Is the requirement incomplete? Should we assume that the client expects a **delete** service to be implemented?
- Regarding the error, could this be a framework-related issue, or is it due to a mistake in my implementation?

I have uploaded a release of my project for you to test the issue: [GitHub Repository - Acme-ANS-D03](#)

Alternatives considered

- **Alternative 1:** I have an incomplete or incorrect implementation of the delete service.
 - **Pros:** The service can be improved, providing a better product for the client and increasing my chances of passing the course.
 - **Cons:** None that I can foresee.
- **Alternative 2:** The issue is caused by a bug in the framework.
 - **Pros:** None.
 - **Cons:** Unlikely, as the framework is rigorously tested.
- **Alternative 3:** Do not implement this service.
 - **Pros:** The error would no longer be an issue.
 - **Cons:** The client may require this feature, and since it is already partially implemented, removing it may not be ideal.

Proposed solution

I have chosen the **first alternative**, so I will continue working on a solution to fix this specific error. Thank you in advance for your time and support.

Best regards,
Gabriel Vacaro

Dear Gabriel María,

Thanks for your detailed analysis regarding the operations that technicians can perform on maintenance records. Thanks, too, for providing a link a release of your project that helps us diagnose the problem and produce a good piece of advice.

Your first question is “is the requirement incomplete? [D03-S05-8] Should we assume that the client expects a **delete** service to be implemented?” Yes, the requirement is incomplete. The customer confirms that a technician who has not published a maintenance record can delete it.

Diagnosing the problem will take a little more time. I’ll keep you posted of my findings as soon as possible.

Regards, RC

Dear Gabriel María,

Unfortunately, I found a problem in your “.project” file. Please, check it:

```
<linkedResources>
  <link>
    <name>acme-framework-25.3.0</name>
    <type>2</type>
    <location>C:/Users/DP2/Desktop/Workspace-25/Frameworks/Acme-
Framework-25.3.0/target/classes</location>    </link>
  <link>
    <name>target/classes</name>
    <type>2</type>
    <location>D:/Courses/DT/24-25/Workspace-25/Projects/Acme-ANS-
D03/target/classes/</location>
  </link>
</linkedResources>
```

We commented on that problem in the tutorial to upgrade your projects. Unfortunately, we don't know the reason why Eclipse transforms some relative paths in the “.project” file into absolute paths that do not work well when the project is migrated to another machine. That didn't happen with previous releases of Eclipse, so it might be a bug in the current release. Unfortunately, we can't report it because we're not able to reproduce it.

I corrected your “.project” file, and I could load your project correctly. Please, correct your local copy as well (just replace the absolute path by a relative path using “WORKSPACE_LOC”). Consult the tutorial on upgrading to the latest version of the workspace for further details.

After correcting your “.project” file, I could load your project and populate the database properly. I then run your project and followed your instructions:

- I logged in as technician1/technician1.
- I requested feature Technician > List my maintenance records on the main menu.
- Unfortunately, the listing does not provide any information on whether a maintenance record is published or unpublished. So, I used the link you provided: <http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/show?id=163>. Note that the link worked because the populator guarantees that the database is populated in a particular order independently from the moment when the population takes place or the machine used. Recall that if you need to add new data and need to preserve this order, then you need to create a new population batch.
- I clicked on the “Tasks” button and no error occurred thus far.
- I then returned to the listing using my browser's return button (not the button in the page)
- I then clicked on the “Delete” button and the browser was redirected to the feature that shows the list of maintenance records, which results in a panic.

The problem shows up when you click on your browser's return button. I'll try to clarify this: the problem's related to controlling navigation within your application; that problem's far from trivial because your application can't easily control the flow of feature requests when the user messes around with the controls provided by the browser. To keep track of the user's navigation, the Acme framework sets a local session in the browser that is called “return-stack”. You can inspect it by pressing F12 to open the developer's tools and then selecting tab “Storage” and then option “Session Storage > <http://localhost:8080>”. Perform the following operations:

- Log in as technician1/technician1.
- Select option “Technician > List my maintenance records” in the main menu.
- Select an unpublished record and then display the session storage for your application.
- There you can see the “return-stack” variable, which must have value “<http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/list?mine=true>”. That is the URL that the “Return” button must request.
- Now click on “Tasks” and display the value of “return-stack” again. Now it must be something like: “<http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/list?mine=true> \n <http://localhost:8080/Acme-ANS-D03/technician/maintenance-record/show?id=170>”. Note that it now has two stacked URLs; if you press the “Return” button, then it returns to the top URL and removes it from the stack; this will take you to the maintenance record form; if you press the

"Return" button in that form, it will take you to the URL at the top of the stack and will empty it; this will take you to the listing of maintenance records.

The conclusion is that as long as you use the "Return" button provided by the application, everything will work quite well because that button co-operated with the other buttons, submits, or menus in your application and will keep the "return-stack" variable properly updated. Unfortunately, if you press the "Return" button in your browser, the "return-stack" variable won't get properly updated, which results in the problem on which you have reported.

If you take a look at the changelog file, you'll easily find out that this is a recurrent problem and the solution provided by the framework is improved little by little, but it's still far from perfect. Anyway, that problem's NOT in your project, but the framework that supports it; thus the problem will be fixed in forthcoming versions. It's not your responsibility to correct it, but we thank you for highlighting this problem.

Regards, RC

PS: please, note that Firefox's developer tools do not always display the values of variable "return-stack" well; Chrome does, but Firefox seems to misinterpret the " \n " used to separate the URLs in the stack in some cases.

Lecciones aprendidas:

- He aprendido la importancia de utilizar siempre los mecanismos de navegación internos de la aplicación para asegurar un correcto control del flujo y evitar errores derivados del uso inadecuado del navegador.
- He comprobado que la gestión adecuada de la pila de retorno (return-stack) es esencial para mantener la coherencia en la navegación y evitar problemas de autorización o redireccionamientos incorrectos.
- He reconocido la necesidad de evitar rutas absolutas en la configuración del proyecto, especialmente en archivos como .project, para garantizar la portabilidad y correcto funcionamiento en distintos entornos de desarrollo.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154 1&n av=discussion board&conf id= 426211 1&forum id= 253522 1&message id= 464651 1

Registro de análisis 5

[D02-S05-R15]

The system must handle technician **dashboards** with the following **indicators**:

- The number of maintenance records grouped by their status.
- The maintenance record with the nearest inspection due date, provided that he or she is involved in any tasks that need to be performed as part of that maintenance.
- The top five aircrafts with higher number of tasks in their maintenance records.
- The average, minimum, maximum, and standard deviation of the estimated cost of their maintenance records in the last year.
- The average, minimum, maximum, and standard deviation of the estimated duration of the tasks in which he or she is involved.

Dear Professor Rafael:

My name is Gabriel Vacaro, and I am a student in group C1.068 for the **Design & Testing II** course. I am reaching out to you because I have a question regarding the implementation of the following requirement:

[D03-S05-R15]

1. The system must handle **technician dashboards** with the following **indicators**:

- The number of maintenance records grouped by their status.
- The maintenance record with the nearest inspection due date, provided that he or she is involved in any tasks that need to be performed as part of that maintenance.
- The top five aircrafts with higher number of tasks in their maintenance records.
- The average, minimum, maximum, and standard deviation of the estimated cost of their maintenance records in the last year.
- The average, minimum, maximum, and standard deviation of the estimated duration of the tasks in which he or she is involved.

Context

Since I have nearly completed the mandatory features D01, D02, and D03, I would like to start working on the supplementary I tasks.

The requirement states that "**The maintenance record with the nearest inspection due date, provided that the technician is involved in any tasks that need to be performed as part of that maintenance.**" However, as has been mentioned several times, tasks are understood to be a catalogue. In fact, a technician can create and publish a task, which can then be used by other members with the same role in their maintenance records. These tasks are simply actions that any technician can perform on aircrafts. For example: [Check the engines], [Inspect the landing gear], [Test the aircraft's electrical system], [Examine the hydraulic systems], [Check the fuel system], [Replace air filters], etc. With this in mind, it is easy to find a maintenance record by inspection due date. However, the part that states "**provided that the technician is involved in any tasks that need to be performed as part of that maintenance**" is unclear. This is because the only technician directly involved in the maintenance record is the one who owns it, not the ones who created and published the tasks that are now being used in that maintenance.

Therefore, my questions are:

Questions

- Is the requirement unclear due to improper elicitation or insufficient specification?
- Does it make sense for the technician dashboard of a specific technician to display maintenance records that do not belong to them?

Alternatives considered

- **A1:** Take into account only the maintenance records that belong to the principal.
 - **Pros:** In the context of Acme-ANS, this is the logical option.
 - **Cons:** No cons identified.
- **A2:** Take into account all the maintenance records that involve tasks created by the principal.
 - **Pros:** No pros identified.
 - **Cons:** This option does not make sense, given the definition of tasks.

- **A3:** Change the relationship between maintenance records and tasks so that the tasks are no longer a catalogue of simple tasks, but rather programmed tasks for the technician who created them to perform when assigned.
 - **Pros:** No pros identified.
 - **Cons:** This option radically opposes the implementation and feedback provided by the lecturers.

Proposed solution

I have chosen the **first alternative**, because it is the only one that is logical in the context of Acme-ANS. Thank you in advance for your time and support.

Gabriel Vacaro
Student#5, Group C1.068
Design & Testing II Course

Related messages:

- [1] https://ev.us.es/webapps/discussionboard/do/message?conf_id= 426211_1&forum_id= 253522_1&course_id= 89154_1&action=list_messages&nav=discussion_board&message_id= 463002_1
- [2] https://ev.us.es/webapps/discussionboard/do/message?conf_id= 426211_1&forum_id= 253522_1&course_id= 89154_1&action=list_messages&nav=discussion_board&message_id= 463680_1
- [3] https://ev.us.es/webapps/discussionboard/do/message?conf_id= 426211_1&forum_id= 253522_1&course_id= 89154_1&action=list_messages&nav=discussion_board&message_id= 461830_1
- [4] https://ev.us.es/webapps/discussionboard/do/message?conf_id= 426211_1&forum_id= 253522_1&course_id= 89154_1&action=list_messages&nav=discussion_board&message_id= 462234_1
- [5] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 463772_1

Dear Gabriel María,

Thanks for reaching out and considering our pieces of advice. Your message is regarding the technician's dashboards, which are a Supplementary I requirement. Please, recall that we strongly recommend that you should start working on your Supplementary requirements when your mandatory requirements are OK, which seems to be your case.

Your question is regarding the requirement that reads as follows:

“ [Show] the maintenance record with the nearest inspection due date, provided that the technician is involved in any tasks that need to be performed as part of that maintenance”.

Clearly, that doesn't make a lot of sense. You correctly mention that tasks are created by individual technicians, but they are included in a global catalogue when they are published. Any technician can use any of the tasks in that catalogue to create his/her own maintenance records.

The requirements elicitation engineer and the customer were contacted and the conclusion is that the last part of the requirement must be deleted. That is, the requirement would read as follows:

“ [Show] the maintenance record with the nearest inspection due date”.

Regarding your alternatives:

- A1: yes, generally speaking, a dashboard shows data regarding a particular user, not global data. You must focus on the maintenance records that were registered by the principal.

- A2: that might be a good interpretation. The requirements engineer likely wrote “involved” instead of “created”. But the customer clarified that it’s only about showing the maintenance record with the nearest inspection date, nothing else.
- A3: no, that’s not correct in this context. The customer has clarified that published tasks form kind of a global catalogue from which technician can pick the tasks involved in their maintenance records.

Regards, RC

Lecciones aprendidas:

- He aprendido que una especificación ambigua puede derivar en interpretaciones erróneas que afectan tanto al diseño como a la experiencia de usuario, por lo que la validación temprana y constante con el cliente es fundamental para evitar retrabajos.
- He valorado la importancia de priorizar la lógica de negocio contextualizada, evitando complicaciones innecesarias, y de buscar la solución más sencilla que cumpla con los objetivos, manteniendo la mantenibilidad y escalabilidad del sistema.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154 1&av=discussion_board&conf_id= 426211 1&forum_id= 253522 1&message_id= 466064 1

Registro de análisis 6

En este caso no se trata de ningún requisito, pero si de un problema relacionado con el workspace:

Regarding "Check-i18n" Script Issue and Configuration Maintainability

Dear Professor Rafael,

My name is Gabriel Vacaro, and I am a student in group C1.068 for the Design & Testing II course.

I am writing to you regarding an issue I encountered with the check-i18n script and to share some reflections on configuration management practices in our Acme projects.

Context

While preparing the release of my Acme-ANS-D04 project, I observed that the script expects the project to use version 25.1.0 of the Acme Framework, as shown in the following line:

```
set "FRAMEWORK=Acme-Framework-25.1.0"
```

However, the current workspace version is 25.4.0. After manually updating this value, the script ran correctly and generated the expected output file.

Nevertheless, I noticed that the report included some inaccurate warnings. For instance, the script flagged the i18n code master.menu.user-account.technician-profile as missing in the English locale, although upon manual verification, I confirmed that the code is correctly defined in the messages-en.i18n file and properly displayed in the application. I have also informed the professors of this issue, and they are already aware of the situation.

Reflections on Configuration Management

This situation led me to reflect on how configuration constants are currently handled within the Acme projects. At present, constants such as the project name, framework version, and default database credentials are either hardcoded in multiple locations or defined symbolically in specific files.

Ideally, it would be preferable to centralise all these constants into a single configuration file, simplifying updates and reducing potential inconsistencies. However, the diverse range of technologies involved in the project (Java, Maven, python, CMD scripts, frontend resources like CSS/JS/HTML, and SQL) makes full centralisation challenging. Additionally, consolidating sensitive information into a single file could pose security risks, especially when publishing the project to platforms like GitHub.

Possible Alternatives

- **A1: Maintain the current approach** but improve documentation and establish stricter guidelines to ensure consistency.
 - Pros: Minimal disruption to existing projects.
 - Cons: Potential risk of inconsistencies over time.
- **A2: Use technology-specific configuration files**, structuring constants appropriately within each context.
 - Pros: Better organisation without excessive technical complexity.
 - Cons: Requires disciplined initial setup and maintenance. May not be quite scalable.
- **A3: Adopt an external configuration management system** using environment variables or secured configuration servers.
 - Pros: High scalability and security.
 - Cons: Significant complexity not justified for projects of this size.

Proposed Solution

Given the context of Acme projects, I believe that adopting technology-specific configuration files (A2) combined with improved documentation (A1) would offer the best balance between maintainability and complexity.

Thank you very much for your time and attention.

Kind regards,
Gabriel Vacaro
Student#5, Group C1.068
Design & Testing II Course

Dear Gabriel María,

Thanks for taking our pieces of advice into consideration. Your question is about the “check-i18n.cmd” companion script. This script performs the following checks on your projects: a) all i18n codes used in your JSP files are defined in all locales, b) many such codes used in your services or validators are also defined, and c) unused codes.

I’ve seen many projects that still show labels like “manager.flight.form.description.label”, which are not properly defined in any i18n files. It’s easy to fix these problems using the “check-i18n” script. Note that the script makes its best at trying to find i18n in your Java code, but it’s difficult to find them all, so, please, review carefully your validators and services to ensure that all i18n codes used there are properly defined.

Thanks for reporting that bug in the script. Yes, the script should have been updated with every release; we updated it internally but forgot to add it to the REVs. I’ve checked that it’s properly packaged into REV4, which will be available on Friday morning.

Your reflection on configuration management is very appropriate. You mention that “constants such as the project name, framework version, and default database credentials are either hardcoded in multiple locations or defined symbolically in specific files”. That’s true and that’s a common problem when you have to integrate multiple technologies: maven to list the supporting components and to define the building process, the JDK to compile your applications, which internally uses a JSP compiler, Eclipse to co-ordinate the work of these tools. And each tool requires different configuration artefacts that do not work together easily.

Your proposal is to collect all of the configuration parameters in a single file, which is our goal but not easy at all. Think of the different technologies involved: how would you make that file available to Maven, JDK, Eclipse, and your own application? Unfortunately, we don't have a simple answer to that question.

Currently, there are only three configuration sources: the ".pom" files, the ".properties" files, and the ".cmd" files; the only duplication, afaik, is the framework version in the ".cmd" and the ".pom" files. The industrial version of the Acme Framework uses an "configurator approach"; a "configurator" is a tool that allows to gather the configuration parameters from an application and set them in a centralised manner. You can issue commands like "configurator show" to list all of the configuration parameters in a project (independently from their exact location/s) or "configurator set parameter value" to change a given parameter (independently from its exact location/s). Unfortunately, this approach is not easy to extrapolate to arbitrary project and it's still under development.

You also mention that the configuration files expose sensitive data like the root credentials, both regarding MariaDB and your own application. A common solution to address that problem is to require developers to define such credentials using environment variables; that is, the credentials are never made explicit in any files. The practical problem with this approach is that many developers create a "setvars" script that sets those variables and sooner or later that script gets stored in a repository.

Unfortunately, there's not a perfect solution to the problems that you mentioned, but we are assessing the configurator approach regarding the configuration parameters and the environment-variable approach regarding the credentials.

Regards, RC

Lecciones aprendidas:

- He aprendido que en proyectos con múltiples tecnologías y herramientas, la gestión centralizada de configuraciones es compleja y requiere un equilibrio cuidadoso entre la mantenibilidad, la seguridad y la practicidad, ya que soluciones simples pueden no ser aplicables sin generar otros problemas.
- Me he dado cuenta de la importancia de detectar y reportar errores en herramientas complementarias (como scripts de chequeo) porque, aunque no sean parte del requisito principal, afectan la calidad y la confianza en el desarrollo, y también fomentan un mejor control de calidad en el proyecto.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154 1&nav=discussion_board&conf_id= 426211 1&forum_id= 253522 1&message_id= 467662 1

Registro de análisis 7

En este caso tampoco se trata de un requisito en específico, sino de un problema detectado en el proyecto Acme-jobs.

[D&T] Regarding "Acme-Jobs-25.5.0" Starter Project Panic Issue with Empty ID Parameters

Dear Professor Rafael,

My name is Gabriel Vacaro, and I am a student in group C1.068 for the Design & Testing II course.

I am writing to you to report a critical issue I have encountered while reviewing the "Acme-Jobs-25.5.0" starter project, specifically concerning how feature requests handle empty id parameters, and to share some reflections on error handling and robustness in service methods.

Context

While testing feature requests in the Acme-Jobs-25.5.0 project, I observed that requesting any feature endpoint (e.g., /employer/job/show?id=) with an empty id parameter immediately causes a panic error, displaying the framework's panic page.

The specific exception thrown is:

```
java.lang.IllegalArgumentException: A null value cannot be assigned to a primitive type
```

Upon analysis, I found that this occurs because the authorise methods in the service classes are written assuming that the id parameter will always be present and correctly formatted as an int. In particular, the methods call `getData("id", int.class)`, directly attempting to map the incoming parameter to a primitive type (int), which cannot accept null values. As a result, when an empty string is provided, the framework fails during type conversion.

Reflections on Error Handling

This situation highlights the importance of designing service layers defensively. When working with user-provided parameters, especially those coming from HTTP requests, it is unsafe to assume that data is always valid. Good practice would require:

- Checking whether the parameter exists and is non-empty before attempting to convert it.
- Using `Integer.class` instead of `int.class` to allow proper handling of null values.
- Failing gracefully (e.g., setting `authorised = false`) rather than triggering a panic.

Possible Alternatives

A1: Maintain the use of primitive types but pre-validate input strictly before attempting type conversion.

- **Pros:** Minimal changes to the service structure.
- **Cons:** Requires careful and repetitive input checking in each service.

A2: Switch from `int` to `Integer` in service methods when fetching request parameters.

- **Pros:** Natural handling of nullability, better error resilience.
- **Cons:** Small technical overhead in adapting the methods accordingly.

Proposed Solution

Given the context and educational goals of the Acme projects, I believe adopting option **B2** — using `Integer` instead of `int` for request data handling— offers the best trade-off between robustness and simplicity.

Final Question

Regarding the project correction during this first call, if this error appears, would it be considered equivalent to failing the project?

Thank you very much for your attention and consideration.

Kind regards,
Gabriel Vacaro
Student#5, Group C1.068
Design & Testing II Course

Dear Gabriel María,

Thanks for taking our pieces of advice into consideration. Your question is regarding the behaviour of an application when it gets an illegal request, which we explained in the last lecture. Please, recall that when you report about a problem in a piece of code it's a good idea to provide a link or a .zip to download the exact project in which that behaviour can be reproduced.

Generally speaking, it should be deauthorised; in most cases, that depends on the requirements, which requires the developer to write the appropriate constraints in an “authorise” method. There are a few cases in which an illegal request can be easily caught because of a “binding problem”, which is something independent from the requirements. There are many other cases in which it requires custom constraints.

For instance, if you hack a request to change an “id” or a “version” into something that is not an integer, or change an enumerated value into something that is not a valid value, then you get a binding exception while executing your methods. For instance, if you write the following statement in an “authorise” method, then it will fail with a binding exception if the user hacks the “id” parameter into something like “XXX”, not an autorisation exception; at least, you’ve not executed the request.

```
id = super.getRequest().getData("id", int.class);
```

Note that this sentence requests to bind the “id” parameter, which must be present, into an “int” value, which is impossible if “id” was hacked into “XXX”.

This problem can be easily addressed by checking the appropriate conditions before invoking such methods. For instance:

```
if (!super.getRequest().hasData("id", int.class))
    ... do not authorise immediately ...
else {
    id = super.getRequest().getData("id", int.class);
    ... check other constraints ...
}
```

But realise that this is quite independent from your requirements. Using this approach will likely result in many similar pieces of code to counter fight trivial hacking attacks. Since this problem is quite independent from the requirements, it argues for some support from your development framework that simplifies it all. There should be a simple means to inform the framework which parameters are required by a particular feature so that it can check them all for binding problems automatically, but that support is not ready yet.

This is the reason why such binding problems are accepted as an outcome from your projects.

Regarding your reflections on the problem:

- “Checking whether the parameter exists and is non-empty before attempting to convert it.” This is a simple approach that works well and makes all constraints (including binding ones) explicit in your code. But a better solution in which you don’t have to write so many binding constraints should exist.
- “Using Integer.class instead of int.class to allow proper handling of null values.” That would handle cases in which the hacker leaves the “id” parameter blank or (maybe) cases in which he or he writes “null” as value; but it would still fail with a binding exception in cases in which the hacker assigns something like “XXX” to the parameter.
- “Failing gracefully (e.g., setting authorised = false) rather than triggering a panic.” I assume that this is not actually a third alternative, but something necessary in your first alternative. If you check that the “id” can’t be converted, then you deauthorise the corresponding request.

Regarding your alternatives, bear in mind that changing your “id” parameter from type “int” into type “Integer” doesn’t solve the problem in cases in which the hacker sets “id” to something that is neither blank not a number.

As a conclusion:

- You may check that the parameters are of the appropriate types before fetching them explicitly using “super.getRequest().getData”; if they are not of appropriate types, then deauthorise the request immediately.
- Think of a better proposal that allows to built those binding constraints into the framework so that the developer doesn’t have to worry a lot about those very common projects. Do you wish to raise your

engineering level; problems like this make a difference amongst engineers who work as developers and engineers who work as R&D engineers.

As we mentioned in the last lecture, we accept a binding exception in such cases, since illegal request with binding problems are not executed. The idea exception would be a deauthorisation exception, but we can accept another exception as long as the framework doesn't provide good support.

Regards, RC

Lecciones aprendidas:

- He aprendido que es fundamental diseñar los métodos de servicio para manejar con defensividad los datos proporcionados por el usuario, especialmente en entornos web, donde parámetros inválidos o vacíos pueden generar excepciones críticas si no se validan previamente.
- Me he dado cuenta de que, aunque cambiar tipos primitivos por sus equivalentes envolventes (por ejemplo, de int a Integer) puede mejorar la robustez ante valores nulos, no es suficiente para evitar todos los errores; es necesario implementar validaciones explícitas y manejar con gracia las solicitudes inválidas para evitar fallos en tiempo de ejecución.

Enlace al foro:

https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussion_board&conf_id=426211_1&forum_id=253522_1&message_id=468491_1

Registro de análisis 8

Requisitos involucrados:

No esta relacionado directamente con un requisito, sino sobre el formal testing en su conjunto.

Regarding Phantom Requests on Formal Testing

Dear Professor Rafael,

My name is Gabriel Vacaro, and I am a student#5 in group C1.068 for the Design & Testing II course.

I am writing to inform you of an issue I have encountered after completing the formal testing phase of my project, specifically regarding the appearance of phantom requests in certain edit features. I would also like to confirm whether this situation is considered acceptable for evaluation purposes.

Context

After conducting formal testing on my project, I noticed the presence of phantom requests in some endpoints. I followed the methodology explained in class to prevent them, and after further investigation, I observed that these phantom requests only occur in .hack test cases, specifically in show methods used within edit features that rely on POST requests.

This behaviour was particularly unexpected, as no phantom requests appear in any .safe test case, nor in any .hack test involving list or show features. The issue only arises in .hack cases tied to edit workflows (i.e., those involving update, create, or publish requests, not delete ones).

Upon analysing the root cause, I found that the phantom requests are not triggered by altering ID values alone, but rather when tampering with the **display text** of enumerated or navigable attributes in <select> inputs. For example:

- In technician/maintenance-record/create, modifying:

```
<option value="59">Registration number: 0000</option>
```

to:

```
<option value="999">HACKED</option>
```

causes a phantom request.

- Similarly, in technician/task/create, modifying the enum field for status from:

```
<option value="0">-----</option>
```

to:

```
<option value="0">HACKED</option>
```

and submitting the form with the remaining fields left empty also results in a phantom request.

This suggests that the phantom is triggered not by the submitted value itself, but by changes to the display text rendered in the browser, particularly in fields tied to enumerations or entity references.

Additionally, I have detected that in one of the .hack test files, an error is thrown due to a navigation attribute or an id field being hacked with a non-integer value. However, I recall that this case was explicitly mentioned in [1] [2] as not being a reason for failure, since the resulting binding exception (e.g., number-format exception) is acceptable under the evaluation criteria.

Question

Since these phantom requests are caused by manipulation of display labels (not the submitted data), and

given that they only appear in .hack files and not during regular usage or .safe tests, I would like to ask whether their presence is acceptable for evaluation purposes, or if I should revise and repeat the affected test cases before proceeding with performance testing.
For reference, I am including a link to the release version of my project where these cases can be reviewed:

[Project Release Link](#)

Thank you very much for your time and attention.

Kind regards,
Gabriel Vacaro
Student #5, Group C1.068
Design & Testing II Course

[1] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253521_1&message_id= 468898_1

[2] https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion_board&conf_id= 426211_1&forum_id= 253522_1&message_id= 468491_1

Dear Gabriel María,

Thanks for taking our pieces of advice into consideration. Your question is about “phantom requests”. Generally speaking, a phantom request is a request that is performed autonomously by your browser or the application itself. The real problem is when they occur before the application and the logging system have started up properly. Such requests typically result in panics and they not recorded properly, which means that they cannot be replayed and that systematically leads to test cases that can’t be repeated. Typically, phantom requests happen when:

1. You hover over a browser tab. Just hovering your mouse over a tab results in an autonomous request to render a thumbnail image in most common browsers. If that request can’t be properly recorded, then it results in disaster.
2. There’s a browser tab in which your application was showing a panic (HTTP GET 500). Many browsers perform phantom autonomous request in an attempt to recover that tab.
3. You perform a request that ends up in panic. In such cases, the framework performs an internal redirection to “any/system/panic”; that is: a single request from the browser results in a phantom request.

Cases (1) and (2) can’t be handled properly because there’s nothing we can do to request Tomcat to refuse calls while your application is starting up: Coyote accepts requests as soon as it can hook on the corresponding HTTP port and Catalina dispatches requests to a particular URL as soon as a controller has registered a service for it. Case (3) is easily handled by the framework since it can keep an eye on every self-request performed by your application.

Regarding your message, you wrote: “Upon analysing the root cause, I found that the phantom requests are not triggered by altering ID values alone, but rather when tampering with the display text of enumerated or navigable attributes in <select> inputs”. Please, note that *that’s not* a phantom request; a phantom request is a request that was performed but was not recorded, which causes the corresponding “.trace” file to be out of sync. The problem shows up typically on banners or other kinds of elements that are generated randomly, as well as when you need to change the current moment: if a trace file is out of sync, then the random banners or the current moment are out of sync and do not match the data expected.

You also wrote that: “Additionally, I have detected that in one of the .hack test files, an error is thrown due to a navigation attribute or an id field being hacked with a non-integer value. However, I recall that this case was explicitly mentioned in [1] [2] as not being a reason for failure, since the resulting binding exception (e.g., number-format exception) is acceptable under the evaluation criteria.” That’s correct: ideally, that should be

de-authorised since that's a clear hacking attempt, but the framework doesn't provide good support for that recurrent problem; so, throwing a format-related exception in such cases will be acceptable.

I downloaded your project, I created the databases, I loaded the project into Eclipse, and I run the sample populator; I didn't find any problems thus far. Then, I run the analyser launcher and I found that your test suite is quite complete, but there are a few missing test cases that you'll be able to identify easily. Please, record additional test cases so that your test suite is as complete as possible.

Then, I run the replay launcher **using the coverage runner**. Most of the test cases run smoothly, except for:

- “technician/misc.hack”: unfortunately, it's not totally clear to me what these test cases are for; please, recall using a feature-oriented approach. Ideally, test cases should be on a per-feature basis, that is: “realm/object/command” or “realm/object”. I assume that your intention was to provide a potpourri of hacking test cases. When I run those test cases, I got a conversion exception since you hacked some ids with values that couldn't be converted into integers; that's acceptable.
- “technician\maintenance-record\publish.hack”: there were a couple of problems:
 - The first one showed up on request “/technician/maintenance-record/show?id=217”. I checked that id 217 does not correspond to a maintenance record, but a task. I could also check that the succeeding requests didn't result in any problems. Furthermore, I could check that the test case expects the request to publish to succeed, since it returns a listing. I run your application, I signed in as “technician1” and I requested the first problematic feature, which resulted in an authorisation failure, as expected. This makes me think that the problem might be that you found a problem in the corresponding service and corrected it, which means that the recorded test cases are not in sync with your implementation: your current implementation works well and results in panic, but the test case was recorded when it resulted in success and returned to the corresponding listing.
 - The next problem showed up on request “/technician/maintenance-record/show?id=219”; I checked that id 219 does not correspond to any entities in your database. I didn't dig too much into this problem because my guess is that it's something similar to the previous one. Please, note that it's a good idea to use numbers like “-1” or “9999” when hacking your ids and navigation attributes, since this facilitates interpreting the test cases.
- “technician\task\publish.hack”: there was only one problem that showed up on request “/technician/task/show?id=217”. I didn't dig too much into this problem, since I think that the cause must be very similar to the previous one.

Thus, my guess is that your test cases do not correspond to the current version of your application, but a previous version in which some illegal requests resulted in success instead of panic. Please, recall our good piece of advice when we started lesson L04: do not record any tests cases or perform any performance testing until you can make sure informally that your features do not have any trivial implementation problem. Informally, you can focus on hacking, which is typically the most usual reason to earn grade “F” (Ley Orgánica 3/2018).

Regards, RC

Dear Professor Rafael,

Thank you very much for your detailed and helpful response.

I would like to clarify a point regarding the id=217 issue that you mentioned in the technician/maintenance-record/publish.hack and technician/task/publish.hack test cases. That identifier corresponds to the first object I created during the test itself. In some hacking test cases, I included the creation of a new maintenance record or task in order to test specific situations.

However, I have confirmed that when I avoid creating new entities and rely exclusively on the existing data from the sample populator, the replayer executes without any issues.

I will make sure to adjust the remaining test cases accordingly and follow the feature-oriented structure as recommended.

Thank you again for your time and guidance.
 Kind regards,
 Gabriel Vacaro
 Student #5, Group C1.068
 Design & Testing II Course

After reviewing the issue more carefully, I have identified the source of the error. The problem occurs when I attempt to manipulate a field by opening the browser's developer tools (using F12). This action unintentionally triggers additional HTTP requests, which are then recorded in the .trace file. For example, if I start the application and access the endpoint /Acme-ANS-D04?debug=true&locale=en, everything behaves as expected, and the generated traces appear normal. However, if I open the developer tools (F12) and inspect the trace again, I observe that the following entries are added:

- "15378992-37eb-47a1-880c-bcdf90013119",GET,"/libraries/datatables/1.10.18/js/jquery.dataTables.min.js",,,200,application/javascript,,8686515223400,8686517497400,8686517982900
- "8d8ab0d7-6342-44e6-9972-99c67cdb6ea5",GET,/any/system/welcome,,,200,text/html;charset=utf-8,,,8686520645799,8686521153399,8686573524400

These additional requests are harmless in this context. However, the situation changes when I log in as technician1 and navigate to /technician/maintenance-record/create. Initially, the form renders correctly. Yet, upon opening the developer tools, the following unauthorised trace entry is appended:

- "37220508-0654-4f38-b44a-2b260afa9c9f",GET,"/technician/maintenance-record/create",,,500,text/html;charset=utf-8,,"Access is not authorised",8767404141800,8767412081900,8767716397200

This indicates that an automatic GET request is triggered upon opening the developer tools, which fails due to access restrictions. Consequently, this unexpected trace pollutes the .trace file and may interfere with test validation.

As a workaround, I suggest opening the developer tools at the very beginning of the testing process and leaving them open throughout. This prevents any unauthorised or redundant requests from being triggered during critical interactions.

That being said, this is only my interpretation of the behaviour. I am not entirely certain, so I kindly request confirmation or correction regarding this hypothesis.

Kind regards,
 Gabriel Vacaro
 Student #5, Group C1.068
 Design & Testing II Course
 Dear Gabriel María,

Thanks for your insight into the problem and trying to clarify it.

It's now clear to me what id "127" referred to; I checked the database after the tests were run, which led me to a wrong conclusion.

Regarding the "phantom requests", please, note that a phantom request is a request that is performed *before* the logging system in your application has initialised; thus, the request is performed and served (typically, commonly, often, but not always, it results in HTTP 500) but it's not stored in the ".trace" file. That causes the traces to be out of sync. If DevTools actually makes a silent request to the system but the logging system is up and running, it'll be recorded and replayed; otherwise, it won't get recorded and the replayer won't work well.

I've digged into the problem that you mention. I downloaded and setup your project again; I started the recorder launcher; then, I started Firefox and made it for the appropriate endpoint (<http://localhost:8082/Acme-ANS-D04?debug=true&locale=en>). I waited for a second to load the application and then checked the log and the trace files (using Sublime Text, which refresh automatically when

the files change). I then opened DevTools, but I couldn't see any changes in the log or the trace file. I opened tab "Network" in DevTools and repeated the process, but I couldn't find any network activity when I pressed F12. The I put a breakpoint in method "TraceLoggerHelper::log", which is used to log messages to the ".trace" files and I repeated the process, but the breakpoint didn't fire when I pressed F12 to open DevTools. I repeated the process again: run the application, open Firefox (with no DevTools), then signed in as "technician1" and made my browser for "/technician/maintenance-record/create"; next, I opened DevTools but I didn't observe any network activity and the breakpoint didn't fire.

Thus, I can't repeat the behaviour on which you comment. In such a situation, I would request you to record a video in which you repeat the steps and describe them as clearly as possible, so that I can see where exactly my problem at reproducing your steps was. Please, show your "Network" tab in DevTools and put a breakpoint at method "TraceLoggerHelper::log", as well.

There are a couple of issues, though. I've decoded the timestamps in the traces that you quoted:

- R1 15378992-37eb-47a1-880c-bcdf90013119: pre-handle-time = 8686515223400 = Mon Apr 07 2245 10:53:43
- R2 8d8ab0d7-6342-44e6-9972-99c67cdb6ea5: pre-handle-time = 8686520645799 = Mon Apr 07 2245 12:24:05
- R3 37220508-0654-4f38-b44a-2b260afa9c9f: pre-handle-time = 8767404141800 = Sat Oct 30 2247 16:02:21

The "pre-handle-time" was studied in the last lecture; it records the moment when a request gets to the system. Note that it is computed using "System.nanoTime()", which does not provide the current wall-clock moment, but relative to an origin that depends on the JVM; thus, the timestamps returned by this method can only be used to compute elapsed times, which is exactly what we need [1]. Thus, don't worry about the previous timestamps referring to years 2245 and 2247. But they are weird since you mention that request R2 was performed after request R1, but the time elapsed is roughly 1.5 hours; and R3 was clearly performed on another run of your application since the year changed by two units (the JVM resets the origin for "System.nanoTime()" when it starts working). Might this be an indication that you have checked the wrong trace files?

Clearly, the workaround that you propose might work: open Firefox, then DevTools, and then make it for your application; that's definitely safe.

Regards, RC

[1] <https://docs.oracle.com/javase/9/docs/api/java/lang/System.html#nanoTime-->

Dear Rafael,

Thank you for your detailed response and for taking the time to analyse the situation so thoroughly. I've already recorded the video as requested; here it is attached [Enlace al video](#). However, I haven't been able to capture what happens in the "Network" tab of DevTools when it first opens. This seems to be precisely the moment when the phantom request is triggered, and since it occurs *upon* opening DevTools, I'm unsure how to record it in time.

I also considered opening DevTools in a different browser beforehand, but I was concerned this might introduce additional phantom requests and distort the result.

Best regards,

Gabriel María

It seems I pasted the wrong link earlier. Here is the correct one that everyone from the US can access: [link](#)

Dear Gabriel María,

Thanks for sharing this video. I've been able to reproduce the behaviour that you described in your initial message.

Indeed, it seems that there's an issue with the "Debugger" tab in Firefox's DevTools implementation. I've been able to check that Firefox makes a GET request for a JS resource and another GET request for the current page when you click on that tab; the first request might succeed, but the second one will definitely not in cases in which it resulted from a POST request or cases where it requires authentication. I repeated the previous procedure several times and no extra requests were made sometimes, but I couldn't characterise the conditions in which that occurs.

It seems more than an issue because the "Network" tab does not record any requests to the application. I've also tried Chrome, which doesn't have a "Debugger" tab, but I've clicked on the other common tabs (sources, console, application, elements, ...) and I didn't find any such problems. Neither does the problem show up with Edge. Thus, my best guess is that this might be a problem with Firefox, which is far beyond our responsibilities. You might well be interested in filing this issue so that Firefox's developers can dig into it and shed a bit of light.

Thus, my suggestion is to use the workaround that you propose: wait for your application to start up, then open Firefox, then open DevTools, then make it for your application. That seems to be quite a safe procedure.

Please note that that's *not* a proper phantom request according to our definition. A phantom request is a request that actually occurs, but is not recorded properly because it happens in between the time Tomcat has started up and Catalina has registered a controller for a particular endpoint and the moment when the logging system is fully initialised. A request to an endpoint for which an endpoint exists before the system is properly initialised results in failures that can't be logged. The request performed when you click on the "Debugger" tab is properly recorded; the problem here is that it's not made by the user that you're recording, but from an anonymous user. Note that the sequence of steps is as follows: – you start your system up – you make Firefox for the welcome page – you sign in as "technician1" – you request feature "technician/maintenance-record/create" – then open DevTools and Firefox makes an *unauthenticated request* to that feature (you can check it by putting a breakpoint at the beginning of the method "AbstractController::handle". The request is properly recorded in the corresponding ".trace" file, but the current version of the recorder can't make a difference between a request that comes from your user interface (which is authenticated as "technician1") and a request that comes from Firefox itself (which is not authenticated). The request from Firefox is properly logged with status "HTTP 500", which indicates failure; but when it's replayed, it's made from the current test case, in which the user is "technician1", which results in status "HTTP 200". Thus, the replayer complains that there's a request that should have failed with status "HTTP 500", but results in success with status "HTTP 200".

Regards, RC

[1] <https://bugzilla.mozilla.org/home>

Dear Professor Rafael,

Thank you very much for your thorough analysis.

I have already reported the issue to the Firefox development team via Bugzilla. If they reply with anything clarifying or useful, I will make sure to share it here in this thread.

Kind regards,

Gabriel Vacaro

Student #5, Group C1.068

Design & Testing II Course

Lecciones aprendidas:

- Se ha comprendido la sensibilidad del testing y la importancia de realizarlo cuando todos los requisitos funcionales hayan sido realizados.
- Se ha aprendido a reportar un bug en Bugzilla.

- Se ha aprendido a usar puntos de ruptura en el código para diagnosticar cuando se graban trazas de ejecución en el fichero *tester.trace*.
- Se ha mejorado la comprensión sobre el *framework* en aspectos de testing y cómo detalles como el navegador web puede introducir inconsistencias muy complejas de diagnosticar y arreglar.

Enlace al foro:

[https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id= 89154_1&nav=discussion board&conf id= 426211_1&forum id= 253522_1&message id= 468988_1](https://ev.us.es/webapps/discussionboard/do/message?action=list_messages&course_id=89154_1&nav=discussion_board&conf_id=426211_1&forum_id=253522_1&message_id=468988_1)

Conclusiones

A lo largo del curso, he demostrado una gran constancia y dedicación, abordando cada desafío con rigor y responsabilidad. He planteado todas las dudas y ambigüedades que han surgido, buscando siempre aclarar cualquier aspecto para evitar malentendidos y errores en la implementación. Gracias a la paciencia y profundidad de las respuestas recibidas, se ha podido mantener un diálogo constructivo que abarcó múltiples temas técnicos y conceptuales.

Esta interacción continuada no solo ha permitido resolver incidencias puntuales, sino que también ha facilitado un aprendizaje más profundo y una mejor comprensión de las buenas prácticas en diseño, testing y desarrollo. Como resultado, el proyecto final ha alcanzado un nivel de calidad elevado, logrando satisfacer las necesidades y requerimientos reales del cliente de manera eficaz y robusta.

Este proceso ha evidenciado que la clave del éxito no solo reside en aplicar conocimientos técnicos, sino en mantener una actitud proactiva, comunicativa y colaborativa durante todo el desarrollo. Esa combinación ha sido fundamental para transformar un conjunto de ideas y requisitos en una solución sólida y profesional, alineada con los objetivos reales del proyecto.

Bibliografía

Web de la universidad de Sevilla - <https://ev.us.es>